Peng Jiang

Phone: 13161911011
Email: jiang-p21@mails.tsinghua.edu.cn
Homepage: https://jp-17.github.io/

Education

Tsinghua University PhD Student in Computational Neuroscience Supervisor: Xiaoxuan Jia **2021.08–present**

Tsinghua University B.Sc. in Life Sciences Minor in Computer Science **2017.08–2021.06**

Research Interests

Keywords: Brain Foundation Model; Brain Encoding & Decoding; World Model; Neural Representation.

Current focus: Large-scale pre-trained models for neural spiking data, neural encoding and decoding, and world models that simulate and understand the world.

Academic Experience

Multi-task Neural Representation Research **2022–2024**

Neural Coding Lab Tsinghua University

- Analyzed multi-task neural representations across brain regions using invasive Neuropixels recordings.
- Built multi-area RNN models to study hierarchical task representation in flexible cognition.
- Studied compositional generalization via continual learning with LoRA fine-tuning of LLMs.

NeuroHorizon: Long-Horizon Forward Prediction of Neural Population Activity via Autoregressive Decoding with Hierarchical Memory **2026**

Peng Jiang, Xiaoxuan Jia · Submitted to NeurIPS 2026 · Under review*

- Studied long-horizon forward prediction of neural population activity.
- Developed an autoregressive encoder-decoder with spike tokenization and hierarchical memory.
- Evaluated multi-step rollout stability across motor- and visual-cortex datasets.

Project Experience

Real-time audio-driven digital human based on 3DGS **2024–2025**

Co-Founder and Algorithm Lead Startup · Leave of absence from PhD

- Took a leave of absence from PhD study to co-found a startup.
- Built a real-time, audio-driven interactive digital human system based on 3D Gaussian Splatting (3DGS).
- Maintained the pipeline for capture calibration, 3DGS reconstruction, and audio-to-expression driving.

NeuroGPT project **2023.06**

Tsinghua University LLM Application Design Challenge, 4th place among 94 teams.